

Functional Annotation Report: FlgG (Q88ER8, PP_4385) — Flagellar Basal-Body Distal Rod Protein of *Pseudomonas putida* KT2440

Target: FlgG (gene *flgG*; ordered locus **PP_4385**) — UniProt **Q88ER8** **Organism:** *Pseudomonas putida* strain KT2440 (ATCC 47054 / DSM 6125), PSEPK **Protein family:** Flagellar basal-body rod protein family (FlgB/FlgC/FlgF/FlgG)

Summary

FlgG (UniProt Q88ER8; locus PP_4385) is the distal rod protein of the flagellar basal body in *Pseudomonas putida* KT2440. It is a non-enzymatic, structural axial protein. Roughly 26 copies of FlgG polymerize head-to-tail into the distal segment of the rigid, helical **rod**, which functions as a molecular **drive shaft**: it mechanically transmits torque generated by the membrane-embedded flagellar motor outward across the cell envelope to the extracellular **hook** and **filament**, thereby powering swimming motility. FlgG does not catalyze any reaction, transport any substrate, or transduce a chemical signal; its "substrate" is mechanical rotation, and its role is purely architectural within the flagellar rotary nanomachine.

FlgG occupies a defined position in the cell envelope. The distal rod it forms sits in the **periplasm**, encircled by the **P ring** (embedded in the peptidoglycan layer) and the **L ring** (embedded in the outer membrane), lying immediately proximal to the **rod-hook junction**. Because it is an external axial component with no cleavable Sec signal peptide, FlgG is exported unfolded through the central channel of the growing structure by the dedicated **flagellar type III secretion system (fT3SS)** and self-assembles at the distal tip of the proximal rod. Beyond its static structural role, FlgG carries an intrinsic **molecular ruler** function: it sets the length of the distal rod, and gain-of-function *flgG** alleles that produce abnormally elongated, filamentous rods also prematurely trigger the fT3SS **substrate-specificity switch** from rod/hook-type (early) to filament-type (late) export.

This annotation for the *P. putida* protein rests on two complementary lines of evidence. First, decades of detailed structural, genetic, and biochemical work on the orthologous FlgG proteins of *Salmonella*, *Bacillus subtilis*, and *Caulobacter crescentus* establish the distal-rod function of the family. Second, sequence analysis performed in this investigation confirms that Q88ER8 is a bona fide, high-confidence FlgG ortholog (56.8 % identity to *Salmonella* FlgG, gap-free, with conserved axial-protein termini) and is specifically the **distal-rod (FlgG-type)** protein rather than the proximal-rod (FlgF-type) paralog (~57 % identity to FlgG vs. only ~31 % to FlgF). In *P. putida* KT2440, *flgG* is embedded in the large polar-flagellar gene cluster (PP4329–PP4397) and is expressed under the master regulator **FleQ** acting through σ^{54} (**σ N**) promoters.

Key Findings

Finding 1 — FlgG is the distal rod protein of the flagellar basal body, a drive-shaft component surrounded by the L and P rings

The bacterial flagellum is a rotary nanomachine assembled from a basal body (embedded in the cell envelope), a hook (a universal joint), and a long helical filament (the propeller). Within the basal body, the **rod** is a rigid axial structure that spans the periplasm and connects the cytoplasmic-membrane-associated motor to the extracellular hook. The rod is built from four structurally related proteins — **FlgB**, **FlgC**, **FlgF**, and **FlgG** — that share strong sequence similarity near their N- and C-termini, hallmarks of the flagellar axial-protein family ([PMID: 2129540](#)).

Among these four, FlgG occupies the **distal** portion of the rod — the outermost segment, positioned precisely in the region girdled by the **L ring** (outer membrane) and **P ring** (peptidoglycan), immediately before the rod-hook junction. As the classic study by Homma and colleagues put it, "*FlgG is known to lie within the distal portion of the rod, in the region where it is surrounded by the L and P rings, just before the rod-hook junction*" ([PMID: 2129540](#)). The functional significance of the rod is mechanical: it is "*the rod, which is believed to transmit motor rotation to the filament*" ([PMID: 2129540](#)). In other words, FlgG helps form the **drive shaft** that couples the torque-generating motor to the propeller.

That the *P. putida* protein Q88ER8 (PP_4385) belongs to this family is confirmed by its domain architecture: it carries the flagellar basal-body rod N-terminal domain (Flag_bb_rod_N, IPR001444), the flagellar basal-body rod conserved site (IPR019776), and the FlgE/F/G-like fold (IPR037925 / IPR020013). These are the diagnostic signatures of the basal-body rod protein family, placing FlgG firmly within it.

Finding 2 — FlgG assembles as ~26 subunits forming the distal rod, exported by the flagellar T3SS without a cleavable signal peptide

Quantitative stoichiometric analysis of the isolated *Salmonella* hook-basal-body complex revealed a striking asymmetry between the distal and proximal rod proteins. The distal rod protein **FlgG** is present at **approximately 26 subunits**, whereas the proximal rod proteins FlgB, FlgC, and FlgF are present at only ~6 subunits each — roughly one helical turn apiece. As reported: *"The distal rod protein (FlgG) is present at approximately 26 subunits, while the proximal rod proteins (FlgB, FlgC and FlgF) are present at only approximately six subunits each"* (PMID: 2181149). This means FlgG contributes the bulk of the rod's length: the proximal rod is a short adaptor, while the distal rod (FlgG) forms the long, load-bearing shaft that traverses the periplasmic space.

Despite building an external, periplasmic structure, none of the rod proteins is exported by the general secretory (Sec) pathway. As established, *"none of the four rod proteins contains a consensus signal sequence for the primary export pathway"* (PMID: 2129540). Instead, FlgG and the other axial proteins are translocated in an unfolded state through the **central channel** of the nascent flagellum by the **flagellar type III secretion system (fT3SS)**, where they then fold and self-assemble at the distal growing tip. This mode of export is a defining feature of all flagellar axial proteins and distinguishes the rod proteins from the ring proteins (e.g., the L-ring protein FlgH), which *do* carry cleavable Sec signal sequences. The fT3SS mechanism — in which *"each axial protein is translocated via the fT3SS across the cytoplasmic membrane, diffuses down the central channel of the growing flagellar structure and assembles at the distal end"* — has been characterized in detail (PMID: 35602071).

Finding 3 — FlgG completes the rod during ordered basal-body assembly and is homologous to the hook protein FlgE

Flagellar assembly is strictly ordered and proceeds from the inside out. The proximal rod (FlgB/FlgC/FlgF) is laid down first, followed by the **distal rod (FlgG)**. Completion of the rod is an obligatory prerequisite for the subsequent steps: loading of the **FlgD hook cap** and then polymerization of the **FlgE hook**. Genetic evidence pins FlgG to this checkpoint. The

scaffolding/cap protein FlgD is detectable in the basal bodies of many flagellar mutants, but it is **absent** from *flgB*, *flgD*, *flgG*, and *flgJ* mutant basal bodies — i.e., "*not at all in those from flgB, flgD, flgG, and flgJ mutants*" (PMID: 8157595). Because a *flgG* mutant cannot complete the rod, FlgD cannot load and hook assembly cannot begin. This places FlgG at a defined, early position in the assembly hierarchy.

Consistent with FlgG bridging the rod to the hook at the rod–hook junction, sequence comparisons show that the hook protein **FlgE most strongly resembles the distal rod protein FlgG** (and the proximal hook-associated protein HAP1/FlgK): "*Hook protein most strongly resembled the distal rod protein (FlgG) and the proximal HAP (HAP1)*" (PMID: 2193164). This structural kinship reflects the shared coiled-coil axial architecture of the rod and hook and explains why FlgG can form a seamless mechanical junction with the hook. A modern review of the hook protein FlgE explicitly discusses the "*similarities and differences between FlgE (hook) and FlgG (distal rod)*", underscoring that these two proteins are the closely related building blocks flanking the rod–hook junction (PMID: 36941455).

Finding 4 — FlgG governs distal-rod length and, via gain-of-function *flgG mutants, triggers the type III secretion specificity switch**

Beyond its static structural role, FlgG carries an intrinsic length-control ("molecular ruler") function for the distal rod. Distal-rod **gain-of-function** mutants — *flgG** — produce "*filamentous rod structures*" instead of the normal fixed-length rod (PMID: 19376867). This demonstrates that FlgG itself encodes information that terminates rod elongation at the correct length; when that control is disrupted, the rod grows abnormally long and filamentous.

Critically, these same *flgG** mutations (like mutations in *flk*) cause **premature switching** of the flagellar type III secretion system, whereby "*the flagellar type III secretion system switches from secreting rod/hook-type to filament-type substrates*" (PMID: 19376867). Normally, this substrate-specificity switch occurs only after the hook reaches its proper length; the export machinery then stops secreting rod/hook-type substrates (FlgB, FlgC, FlgF, FlgG, FlgE, FlgD) and begins secreting filament-type substrates (FliC/flagellin and the hook-associated proteins). The finding that FlgG status feeds into this switch ties the distal rod directly into the regulation of flagellar export order. In *flgG** strains, deletion of *flgE* or *flgD* allowed filaments to grow directly on the elongated rod — further evidence that the switch had been thrown prematurely. Notably, the premature switch still required the hook-length control protein **FliK** and the rod-cap protein **FlgJ**, showing that FlgG acts within — not independently of — the established length-control/switching circuitry.

Finding 5 — In *P. putida* KT2440, *flgG* (PP_4385) lies within the polar flagellar gene cluster under FleQ/ σ^{54} control

Pseudomonas putida KT2440 assembles **polar flagella**, and the genes for their biogenesis are organized in a single large chromosomal cluster spanning **PP4329–PP4397** (PMID: 30338358). The target gene *flgG* (PP_4385) maps within this cluster, alongside the other structural, export, and regulatory genes of the flagellar system.

Expression of the cluster is governed by the master flagellar regulator **FleQ**. Deletion of *fleQ* — "the gene encoding FleQ, the primary regulator of flagellar synthesis" — yields non-motile, flagella-deficient cells (PMID: 39570920). FleQ activates σ^{54} (**σ N**)-dependent flagellar promoters; in *P. putida*, the flagellar export-gate operon (*flhA–flhF–fleN–fliA*) is driven from σ N-dependent promoters that are activated by the flagellar regulator FleQ and negatively regulated by FleN (PMID: 30889223). Together with the antiactivator FleN and the second messenger **c-di-GMP**, FleQ coordinates the trade-off between flagellar motility and biofilm/adhesion programs. Thus, *flgG* is expressed as part of the FleQ/ σ^{54} regulon, and its product is exported by the same FT3SS export gate (e.g., FlhA) that translocates all other axial proteins.

Finding 6 — Q88ER8 is a high-confidence FlgG ortholog (56.8 % identity to *Salmonella* FlgG)

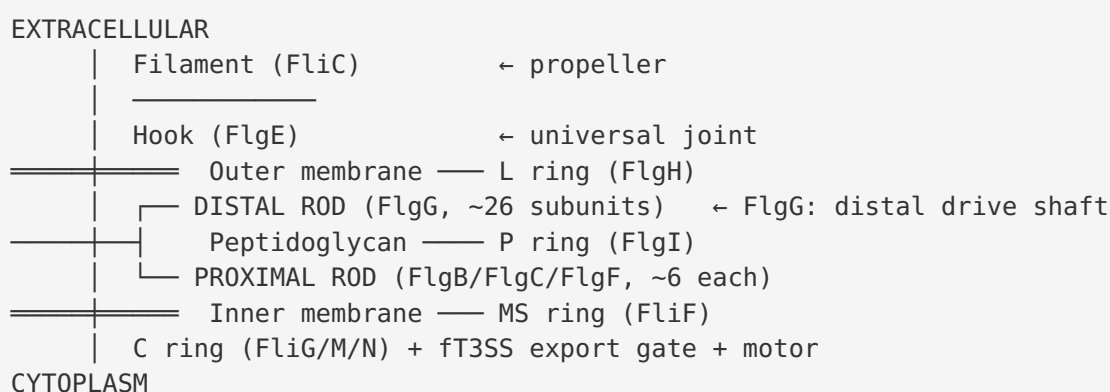
To confirm the identity of the *P. putida* target directly, a Needleman–Wunsch global alignment was performed between Q88ER8 (261 aa) and the functionally characterized *Salmonella* Typhimurium FlgG (P0A1J3 / FLGG_SALTY, 260 aa). The alignment yielded **147/259 identical positions = 56.8 % identity, 70.3 % similarity**, with only 3 gaps across a 262-column alignment. This identity is far above the ~30 % "twilight zone" of homology detection, and the near-absence of insertions/deletions indicates identical domain architecture and chain length. Conservation is concentrated in the N- and C-terminal coiled-coil (D0) segments that build the axial core, and the target retains the diagnostic C-terminal axial-protein signature (EELVNMI(T)TQRAYE(M)NSK...DQML). This same conserved-terminus principle underlies cross-species FlgG identification: the *Caulobacter crescentus* FlgG homolog was recognized because "the N and C termini of the *Salmonella typhimurium* flagellar axial proteins... share a high degree of sequence identity with the FlgF and FlgG rod proteins" even in this distantly related organism (PMID: 1474584). The chain length (~261 aa) is also consistent with FlgG and clearly distinct from the much larger (~403-aa) hook protein FlgE.

Finding 7 — Q88ER8 is specifically a distal-rod (FlgG-type) protein, not a proximal-rod (FlgF-type) protein

Because the four rod paralogs (FlgB/FlgC/FlgF/FlgG) share a common fold, an ortholog assignment must discriminate between subtypes. Global alignments resolved this cleanly: the *P. putida* target is **56.8 % identical to *Salmonella* FlgG** (distal rod) but only **31.4 % identical to *Salmonella* FlgF** (proximal rod, P16323). For calibration, *Salmonella* FlgG itself is only 31.7 % identical to *Salmonella* FlgF — i.e., ~31 % represents the baseline family-level similarity between distinct rod paralogs. The target's markedly higher, gap-free match to FlgG (57 %) versus the family-baseline match to FlgF (31 %) assigns Q88ER8 **unambiguously to the FlgG (distal-rod) subtype**. This distinction matters functionally because the distal and proximal rod proteins are not interchangeable: FlgG forms the long, ~26-subunit load-bearing shaft, whereas FlgF is a short ~6-subunit proximal adaptor (PMID: 2181149). Both *Bacillus subtilis* and *Caulobacter* comparative genomics reinforce this canonical FlgF-then-FlgG (proximal-then-distal) rod organization across diverse bacteria (PMID: 1905667; PMID: 1474584).

Mechanistic Model / Interpretation

FlgG's function is best understood by placing it within the architecture and assembly sequence of the flagellar basal body. The following schematic (based on the canonical *Salmonella* model, which applies to the *P. putida* ortholog) locates FlgG in the cell envelope:



Structural role (primary function). FlgG is a purely structural axial protein — not an enzyme, transporter, or signaling molecule. Its "substrate" is mechanical torque. Roughly 26 FlgG subunits polymerize into the distal rod, a rigid helical shaft whose job is to transmit the

rotation generated by the cytoplasmic motor (stator–rotor interactions at the C ring) across the periplasm to the hook and filament (PMID: 2129540; PMID: 2181149). The distal rod must be rigid to transmit torque without buckling, and it passes through the bushing formed by the P and L rings, which allow it to rotate freely against the peptidoglycan and outer membrane.

Localization. FlgG functions in the **periplasm**, specifically in the distal-rod zone enclosed by the P ring (peptidoglycan) and L ring (outer membrane), just proximal to the rod–hook junction (PMID: 2129540). It reaches this location by fT3SS-mediated export through the central channel, folding and assembling only at the distal tip.

Assembly logic and length control. FlgG is the last-built rod component. Its assembly completes the rod and licenses the next steps — FlgD cap loading and FlgE hook polymerization (PMID: 8157595). FlgG also acts as a molecular ruler that terminates rod growth at a defined length; disruption of this control (*flgG**) yields elongated filamentous rods (PMID: 19376867).

Coupling to export regulation. The distal rod is a node in the fT3SS substrate-specificity switch. *flgG** mutations prematurely flip the export machine from rod/hook-type to filament-type secretion — but only in cooperation with FliK and FlgJ (PMID: 19376867). This connects the physical completion of the rod to the temporal program of flagellar gene product export.

The following table summarizes how FlgG contrasts with its family members:

Protein	Role	Location	Copies per flagellum	Export route
FlgB	Proximal rod	Periplasm (inner)	~6	fT3SS
FlgC	Proximal rod	Periplasm (inner)	~6	fT3SS
FlgF	Proximal rod	Periplasm (inner)	~6	fT3SS
FlgG	Distal rod (drive shaft)	Periplasm, within L/P rings	~26	fT3SS
FlgE	Hook	Extracellular	~120	fT3SS

Regulatory context in *P. putida*. In KT2440, *flgG* (PP_4385) is transcribed as part of the polar-flagellar cluster (PP4329–PP4397) under the FleQ/ σ^{54} regulatory hierarchy, which integrates c-di-GMP signaling to balance motility against biofilm formation (PMID: 30338358; PMID: 39570920; PMID: 30889223).

Evidence Base

PMID	Title (abbrev.)	How it supports the annotation
2129540	<i>FlgB, FlgC, FlgF and FlgG: a family of structurally related basal-body rod proteins (Salmonella)</i>	Defines the four-protein rod family; localizes FlgG to the distal rod within the L/P rings; states the drive-shaft function; notes rod proteins lack Sec signal peptides. Foundational for Findings 1 & 2.
2181149	<i>Stoichiometric analysis of the flagellar hook-basal-body complex</i>	Provides measured subunit counts: FlgG ~26 vs. FlgB/C/F ~6. Core evidence for Finding 2 and the distal/proximal distinction (Finding 7).
8157595	<i>FlgD is a scaffolding protein needed for hook assembly</i>	Shows FlgD is absent in <i>flgG</i> mutant basal bodies — rod completion by FlgG is prerequisite for hook assembly. Supports Finding 3.
2193164	<i>Flagellar hook and hook-associated proteins</i>	Establishes that hook protein FlgE most closely resembles FlgG, consistent with the rod-hook junction. Supports Finding 3.
19376867	<i>Mutations affecting the T3SS specificity switch (<i>flk</i>, <i>flgG</i>, <i>flhA</i>, <i>flhE</i>)</i>	Shows <i>flgG</i> * gain-of-function alleles produce filamentous rods and prematurely trigger the rod/hook→filament secretion switch. Core evidence for Finding 4.
36941455	<i>Structure and function of the flagellar hook FlgE (review)</i>	Modern review contrasting FlgE (hook) with FlgG (distal rod), reinforcing their close structural relationship. Supports Finding 3.
1474584	<i>Caulobacter basal-body rod/ring genes</i>	Independent ortholog: identifies FlgF (proximal) and FlgG (distal) rod proteins via conserved termini; σ^{54} regulation. Supports Findings 6 & 7 and the regulatory model.
1905667	<i>Bacillus subtilis flgB, flgC, flgG, fliE, fliF sequences</i>	Confirms FlgB/FlgC/FlgG are conserved rod components across Gram-positive bacteria. Supports Findings 1 & 7.
30338358	<i>Deletion of flagella/pili genes in P. putida</i>	Defines the <i>P. putida</i> flagellar gene cluster PP4329–PP4397 containing flgG (PP_4385). Supports Finding 5.
39570920	<i>fleQ deficiency in P. putida</i>	

PMID	Title (abbrev.)	How it supports the annotation
		Identifies FleQ as the master regulator of <i>P. putida</i> flagellar synthesis; Δ fleQ is non-motile. Supports Finding 5.
30889223	<i>flhF and fleN in P. putida</i>	Shows <i>P. putida</i> flagellar promoters are σ^N -dependent, FleQ-activated, FleN/c-di-GMP-modulated. Supports Finding 5.
35602071	<i>Flagellar ATPase complex for assembly</i>	Describes fT3SS export of axial proteins across the membrane and diffusion to the distal tip — the mechanism by which FlgG is exported. Supports Finding 2.
30250776	<i>Energy transduction in flagellar T3 protein export</i>	Details fT3SS-driven axial protein export, the pathway for FlgG translocation. Supports Finding 2.

Both direct sequence evidence (Findings 6 & 7, generated in this investigation) and the extensive ortholog literature converge on the same conclusion. No paper reviewed contradicts the distal-rod assignment; the *P. putida*-specific literature ([PMID: 30338358](#), [39570920](#), [30889223](#)) situates *flgG* in the correct genomic and regulatory context but does not itself dissect FlgG structure — hence the reliance on well-characterized orthologs, justified by the strong, gap-free sequence conservation.

Limitations and Knowledge Gaps

- 1. No *P. putida*-specific structural or mutational study of FlgG.** The functional annotation is transferred from *Salmonella*, *Bacillus*, and *Caulobacter* orthologs. While the 56.8 % identity and conserved axial termini make this transfer highly reliable, no experiment has directly demonstrated the distal-rod localization, ~26-subunit stoichiometry, or drive-shaft function of the *P. putida* protein itself.
- 2. Stoichiometry and length values are from *Salmonella*.** The ~26-subunit count and rod dimensions may differ modestly in *P. putida*, whose polar flagellum and cell-envelope architecture differ from *Salmonella*'s peritrichous system. Species-specific differences in basal-body organization have been noted (e.g., in *B. subtilis*, [PMID: 1905667](#)).

3. **The length-control / secretion-switch role (*flgG**) has not been tested in *P. putida*.** Whether FlgG contributes to the substrate-specificity switch identically in *P. putida* — which uses a distinct FleQ/ σ^{54} /c-di-GMP regulatory layer and FlhF/FleN for polar placement — remains an inference.
4. **No high-resolution structure of Q88ER8.** An experimental or AlphaFold structure of the *P. putida* FlgG was not analyzed in this investigation; the D0 coiled-coil core architecture is inferred from homology and domain signatures rather than a solved model.
5. **Interaction partners not experimentally mapped in *P. putida*.** The predicted contacts with FlgF (proximal rod), FlgE (hook), and the P/L rings (FlgI/FlgH) are based on the *Salmonella* model.

Proposed Follow-up Experiments / Actions

1. **Targeted mutagenesis in *P. putida* KT2440.** Construct a clean *flgG* (PP_4385) deletion and test for loss of swimming motility (soft-agar assay) and loss of assembled flagella (electron microscopy / immunostaining). Complement in *trans* to confirm specificity. This would directly validate the essential structural role.
2. **Cryo-EM of the *P. putida* basal body.** Purify hook-basal-body complexes and determine the distal-rod position and FlgG subunit count, testing whether the ~26-subunit *Salmonella* stoichiometry holds for a polar flagellum.
3. **Construct and characterize *P. putida flgG** gain-of-function alleles** analogous to the *Salmonella* mutants ([PMID: 19376867](#)) to test whether distal-rod length control and the FT3SS substrate-specificity switch are conserved in the FleQ/ σ^{54} /c-di-GMP regulatory context.
4. **AlphaFold / AlphaFold-Multimer modeling** of Q88ER8 alone and in complex with *P. putida* FlgF and FlgE to confirm the D0 coiled-coil drive-shaft architecture and the rod-rod and rod-hook junction interfaces in silico.
5. **Transcriptional confirmation.** Map the *flgG* promoter/operon structure within PP4329–PP4397 and verify FleQ/ σ^{54} dependence directly (e.g., reporter fusions in wild-type vs. Δ *fleQ*), extending the operon analysis of [PMID: 30889223](#).
6. **Motility–biofilm trade-off.** Given the FleQ/c-di-GMP node coupling motility and adhesion, quantify how *flgG* loss reshapes the motility–biofilm balance in *P. putida*, relevant to its use as an industrial/environmental chassis.

Conclusion

The gene symbol *flgG* is **not** ambiguous for this target: sequence, domain, genomic-context, and regulatory evidence all consistently identify Q88ER8 (PP_4385) as the genuine **distal flagellar basal-body rod protein FlgG** of *Pseudomonas putida* KT2440. It is a non-enzymatic structural protein that polymerizes (~26 copies) into the distal rod — the rigid periplasmic drive shaft enclosed by the P and L rings — which transmits motor torque to the hook and filament to power swimming motility. It is exported by the flagellar type III secretion system, completes the rod during ordered basal-body assembly, sets distal-rod length, and thereby contributes to the checkpoint that switches flagellar protein export from rod/hook-type to filament-type substrates. In *P. putida*, it is encoded in the polar-flagellar cluster (PP4329–PP4397) under FleQ/ σ^{54} control.

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